

Qur'an Learning Platform

Deployment Architecture & Cost Comparison

Recommended hosting setup for the prototype phase (up to 200 users), with a transparent, component-by-component cost comparison between self-hosting and managed alternatives.

Prepared for client review
August 28, 2026

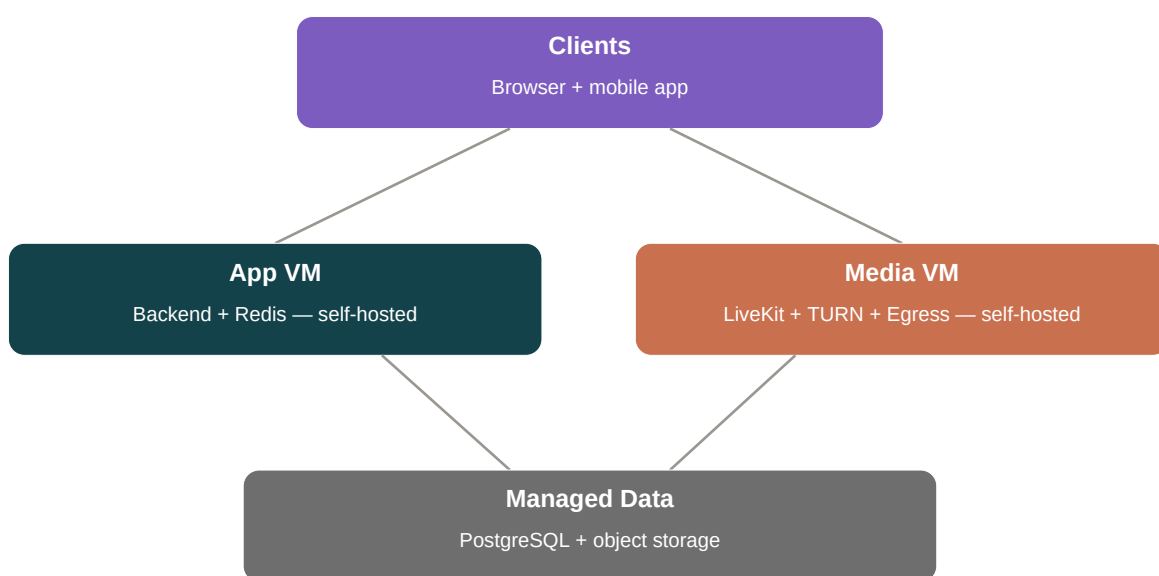
1. Purpose & Scope

This document covers the deployment architecture recommended for the platform's prototype phase — up to 200 users — and a transparent cost comparison between self-hosting each infrastructure component and using a managed equivalent. It complements the architecture and delivery plan already reviewed, and focuses specifically on hosting decisions and their true cost, including engineering time.

Decision already made for this phase: the application backend and the video/call layer are self-hosted; the database is managed, specifically to protect against data loss during early testing.

2. Deployment Architecture

The prototype deployment is intentionally minimal: two small servers and two managed data services. No container orchestration, no multi-region setup, and no autoscaling — all realistic future needs, but not needed at this scale.



- **App VM** — runs the backend application and a small Redis instance, used for caching, live presence, and as the internal job queue for recording tasks (see below).
- **Media VM** — runs the self-hosted LiveKit media server, a TURN server for participants on restrictive networks, and the Egress recording service. It has its own public IP and is kept separate from the App VM for a cleaner security boundary around the server that must expose open ports to the internet.
- **Managed PostgreSQL** — the platform's primary database, hosted by a managed provider rather than self-hosted, so backups and failure recovery are handled automatically during the pilot.
- **Managed object storage** — an S3-compatible service for recordings and uploaded lesson videos.

Implementation note: LiveKit's Egress recording service uses Redis as its internal job queue, even on a single node. Rather than running a second Redis instance, the Media VM's Egress service points at the same Redis already running on the App VM — one shared dependency, not a new one.

3. How This Comparison Is Built

- Self-hosted figures include three parts: monthly server cost, one-time setup time (already part of Sprint 1's infrastructure work in the delivery plan — not an additional line item), and ongoing monthly maintenance time.
- Labor is costed at €13/hour, as provided.
- Only the labor that differs between the two approaches is counted. Writing the application code that talks to the video layer or the database is the same work either way, so it is not duplicated in this comparison — only the infrastructure-specific setup (servers, TURN, TLS, container config) is unique to self-hosting.
- Time estimates assume an experienced engineer with no major blockers — a light, realistic case, not a padded worst case.
- Server and managed-service prices reflect publicly published pricing as of August 2026 (Hetzner Cloud, DigitalOcean Managed Databases, LiveKit Cloud), converted to EUR where quoted in USD. Confirm exact figures at time of procurement — cloud pricing changes.

4. Component-by-Component Comparison

Component	Self-Hosted	Managed Alternative
App Hosting	Small VM (2 vCPU / 4 GB): ≈€6/mo Setup: 4h ≈ €52 (one-time, in Sprint 1) Maintenance: 1h/mo ≈ €13/mo	Small managed container platform ≈€20/mo Setup & maintenance: negligible
Video / Real-Time (LiveKit)	VM (4 vCPU / 8 GB) + TURN + Egress: ≈€11/mo Setup: 5h ≈ €65 (one-time, in Sprint 1) Maintenance: 1h/mo ≈ €13/mo	LiveKit Cloud €0 (free tier, light testing) to ≈€46/mo (real pilot use) plus recording fees if used heavily
Database	Not used — ruled out to protect pilot data from loss	Managed PostgreSQL (small instance) ≈€15/mo
Object Storage	Not practical at this scale	S3-compatible object storage ≈€5/mo at light usage

5. Total Monthly Cost

Steady-state monthly cost, once Sprint 1 setup is complete — infrastructure plus the ongoing maintenance time it takes to run it:

	Monthly Infra	Monthly Maintenance Labor	Total / Month
This architecture (App + Video self-hosted, DB + storage managed)	≈€37 (€6 + €11 VMs, €15 DB, €5 storage)	≈€26 (2h × €13)	≈ €63/mo
Fully managed alternative (everything managed)	≈€40–€86 (PaaS + Cloud + DB + storage)	Negligible	≈ €40–€86/mo

One-time setup, already covered within Sprint 1 of the delivery plan, shown here for transparency: 9 hours ≈ €117.

6. Why Self-Host at This Scale

At 200 users, the two paths land in a similar monthly range — self-hosting is not dramatically cheaper yet. Its value here is structural rather than immediate: infrastructure cost stays fixed as usage grows toward what these two servers can actually handle — hundreds of concurrent participants — while the managed path's cost is metered and grows with every additional minute of video used. The gap that matters opens up exactly when the platform succeeds and usage increases, which is the scenario worth planning for now rather than after the fact. Self-hosting also means the data stays on infrastructure this project controls from day one, and the setup labor is already part of Sprint 1 — not a separate cost to weigh against the client's budget.

7. Assumptions & Caveats

- Figures reflect publicly published pricing as of August 2026 and an approximate USD→EUR conversion; reconfirm exact numbers before procurement.
- Assumes light-to-moderate pilot usage across 200 total users, not 200 concurrent video participants.
- The specific VM provider used for pricing is illustrative of typical small-server cost; the final provider and region should follow the data-residency decision raised in the architecture plan.
- Labor estimates assume an experienced engineer with no major blockers; real-world hours can vary with unforeseen issues.